AP BIOLOGY TOPIC 3.5

Cellular Respiration: Glycolysis, Krebs Cycle, Oxidative Phosphorylation & Fermentation

UNIT 3: CELLULAR ENERGETICS (12–16% EXAM WEIGHT)

ENE-1.K | ENE-1.L | ETC, CHEMIOSMOSIS & FERMENTATION

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 3.5 (CELLULAR RESPIRATION: GLYCOLYSIS, KREBS CYCLE, OXIDATIVE PHOSPHORYLATION & FERMENTATION)

CED Core Vocabulary

Overall Exergonic Equation — $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + \sim 30–32$ ATP + Heat ($\Delta G = -686$ kcal/mol) Glucose ($C_6H_{12}O_6$) is oxidized to 6 CO_2 ; Oxygen (O_2) is reduced to 6 H_2O .

Macromolecules as Energy Sources — Carbohydrates: Primary, immediate fuel source (~4 kcal/g); glucose enters glycolysis directly.

Lipids (Fats) — Long-term energy reserve; abundant reduced C–H bonds yield >2× more ATP per gram (~9 kcal/g) via beta-oxidation into Acetyl-CoA.

Proteins — Used only during prolonged starvation after deamination (removing -NH_2 as urea).

Outer Membrane — Smooth boundary separating mitochondrion from cytosol.

Inner Membrane (Cristae) — Highly folded infoldings (cristae) that dramatically increase surface area to volume ratio to embed thousands of copies of ETC protein complexes and ATP Synthase.

Intermembrane Space — Narrow space between membranes where H^+ ions are pumped (low pH / acidic).

Mitochondrial Matrix — Innermost fluid compartment (higher pH); site of Pyruvate Oxidation and the Krebs Cycle.

§2. OVERVIEW, MACROMOLECULE FUELS & MITOCHONDRIA

Exergonic Redox

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

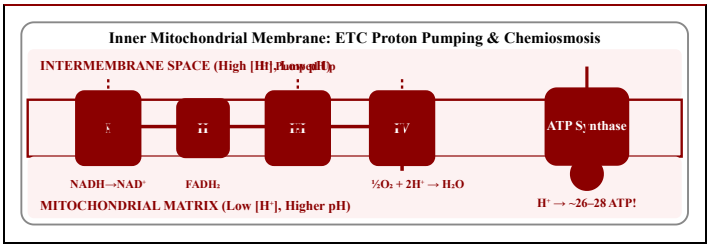
Overview, Macromolecule Fuels & Mitochondria ➔ **Overall Exergonic Equation** ➔ **Macromolecules as Energy Sources**

- **Overall Exergonic Equation:**
  $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + \sim 30–32$ ATP + Heat (**$\Delta G = -686$ kcal/mol**)
  - **Glucose ($C_6H_{12}O_6$) is oxidized** to **6 CO_2** ; **Oxygen (O_2) is reduced** to **6 H_2O** .
- **Macromolecules as Energy Sources:**
  - **Carbohydrates:** Primary, immediate fuel source (**~4 kcal/g**); glucose enters glycolysis directly.
  - **Lipids (Fats):** Long-term energy reserve; abundant reduced C–H bonds yield **>2× more ATP per gram (~9 kcal/g)** via beta-oxidation into Acetyl-CoA.
  - **Proteins:** Used only during prolonged starvation after deamination (removing **-NH_2** as urea).
- **Mitochondrial Double-Membrane Structure → Function:**
  - **Outer Membrane:** Smooth boundary separating mitochondrion from cytosol.
  - **Inner Membrane (Cristae):** Highly folded infoldings (**cristae**) that dramatically **increase surface area to volume ratio** to embed thousands of copies of ETC protein complexes and ATP Synthase.
  - **Intermembrane Space:** Narrow space between membranes where **H^+** ions are pumped (**low pH / acidic**).
  - **Mitochondrial Matrix:** Innermost fluid compartment (**higher pH**); site of Pyruvate Oxidation and the Krebs Cycle.

§4. STAGE 4: OXIDATIVE PHOSPHORYLATION (ETC & CHEMIOSMOSIS)

Inner Membrane Cristae

- **Electron Transport Chain (ETC — Complexes I–IV):**
  - **NADH** (at Complex I) and **$FADH_2$** (at Complex II) are **oxidized**, donating high-energy electrons harvested from food.
  - Electrons step down a chain of increasingly electronegative redox carriers, releasing free energy used by Complexes I, III, and IV to actively pump **H^+ from the matrix into the intermembrane space**.
  - **Final Electron Acceptor = Oxygen (O_2):** Highly electronegative **O_2** pulls electrons at the bottom of the chain and combines with matrix protons to form **water ($\frac{1}{2}\ O_2 + 2e^- + 2H^+ \rightarrow H_2O$)**. Without **O_2**, the entire ETC backs up and halts!
- **Proton Gradient & Chemiosmosis (ATP Synthase):**
  - **Intermembrane Space:** High **$[H^+]$** → **Low pH (more acidic)**.
  - **Mitochondrial Matrix:** Low **$[H^+]$** → **Higher pH (more basic)**.
  - **H^+** ions flow back down their electrochemical gradient into the matrix through the rotary **ATP Synthase** turbine, driving **oxidative phosphorylation** (**$ADP + P_i \rightarrow \sim 26–28$ ATP**).
  - **Uncoupling (Thermogenesis):** Uncoupler proteins (e.g., **UCP1 / thermogenin** in hibernating mammal brown fat) create **H^+** leak channels across the inner membrane → dissipates the proton gradient as **body heat** instead of making ATP!



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§3. STAGES 1–3: GLYCOLYSIS, LINK REACTION & KREBS CYCLE

Harvesting NADH & $FADH_2$

- **Stage 1 — Glycolysis (Cytosol | Anaerobic — No O_2 Needed):**
  - Splits 1 **Glucose (6C) → 2 Pyruvate (3C)** in a 10-step enzyme pathway conserved across all domains of life.
  - **Energy Investment (2 ATP used) vs. Payoff (4 ATP made):** Yields **2 Net ATP** (via substrate-level phosphorylation) + **2 NADH** (from **2 NAD^+**) + **0 CO_2**.
- **Stage 2 — Pyruvate Oxidation / Link Reaction (Mitochondrial Matrix):**
  - Each **Pyruvate (3C)** is actively transported into the matrix, loses a carboxyl group as **CO_2**, reduces **$NAD^+ \rightarrow NADH$**, and attaches to Coenzyme A to form **Acetyl-CoA (2C)**.
  - **Yield per Glucose (2 Pyruvates): 2 Acetyl-CoA + 2 CO_2 + 2 NADH** (0 ATP).
- **Stage 3 — Krebs / Citric Acid Cycle (Mitochondrial Matrix):**
  - Completes the oxidation of organic carbon: **Acetyl-CoA (2C)** joins oxaloacetate (4C) to form citrate (6C), which is oxidized stepwise back to 4C.
  - **Yield per Glucose (2 turns): 4 CO_2 + 2 ATP (GTP) + 6 NADH + 2 $FADH_2$**.
  - **Main Purpose:** Generate loaded electron carriers (**NADH** & **$FADH_2$**) to feed the ETC!

[COMMON AP EXAM FRQ TRAP] Two Classic Exam Mistakes:

1. **Glycolysis makes 4 Gross ATP, but only 2 NET ATP** per glucose.
2. **Most ATP is NOT made in the Krebs cycle** (only 2 ATP!) — ~90% of ATP (**~26–28 ATP**) is made in **Oxidative Phosphorylation**.

§5. STAGE SUMMARY TABLE & ANAEROBIC FERMENTATION

Regenerating NAD^+

STAGE	LOCATION	INPUTS (PER GLUCOSE)	NET OUTPUTS (PER GLUCOSE)
1. Glycolysis	Cytosol	1 Glucose (6C), 2 **NAD^+**, 2 ATP	**2 Pyruvate, 2 Net ATP, 2 NADH**
2. Pyruvate Oxid.	Matrix	2 Pyruvate (3C), 2 **NAD^+**, CoA	**2 Acetyl-CoA, 2 CO_2, 2 NADH**
3. Krebs Cycle	Matrix	2 Acetyl-CoA (2C), 6 **NAD^+**, 2 FAD	**4 CO_2, 2 ATP, 6 NADH, 2 $FADH_2$**
4. Ox. Phosphor.	Inner Memb.	10 NADH, 2 **$FADH_2$**, 6 **O_2**	**~26–28 ATP, 6 H_2O**, 10 **NAD^+**, 2 FAD

- **Anaerobic Fermentation (Cytosol — No O_2 Required | Yield = 2 ATP):**
  - **Sole Purpose:** Oxidize **$NADH \rightarrow NAD^+$** by transferring electrons to pyruvate (or its derivative) so **glycolysis does not run out of NAD^+** and can continue making **2 ATP per glucose**! Fermentation steps themselves produce **0 additional ATP**.
  - **Lactic Acid Fermentation:** Pyruvate is directly reduced by NADH → **Lactate (lactic acid, 3C) + NAD^+** (no **CO_2** released; occurs in human skeletal muscle & yogurt bacteria).
  - **Alcohol Fermentation:** Pyruvate → acetaldehyde + **CO_2** → reduced by NADH to **Ethanol (2C) + NAD^+** (occurs in yeast & some bacteria).

[MECHANISTIC NOTE] AP Exam Scope Note: You do NOT need to memorize individual intermediate names of glycolysis/Krebs or specific enzyme/carrier names — focus on **compartment locations, H^+ gradient pH directions, inputs/outputs, and why O_2 or fermentation is required**.

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 3.5 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 3.5: Cellular Respiration: Glycolysis, Krebs Cycle, Oxidative Phosphorylation & Fermentation ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 5, 6 & 13 (Enzyme Catalysis, Photosynthesis Leaf Disks & Respirometry): Enzyme assay (catalase/peroxidase spectrophotometry), Floating Leaf Disk Assay (ET_{50} = time for 50% of disks to float in $NaHCO_3$), and KOH Respirometry (KOH absorbs CO_2 gas as a solid precipitate so volume decrease directly measures O_2 consumption). Negative Control: boiled enzyme/boiled seeds or glass beads (corrects for ambient temp/pressure changes). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Reaction Rate = ΔProduct / ΔTime (initial linear slope V_0). Michaelis-Menten Kinetics: Competitive inhibitors increase apparent K_m without changing V_{max}; Noncompetitive/allosteric inhibitors decrease V_{max} without changing K_m. Gibbs Free Energy: $\Delta G < 0$ (exergonic/spontaneous, coupled via ATP → ADP + P_i, $\Delta G^{\circ\prime} = -7.3$ kcal/mol) vs. $\Delta G > 0$ (endergonic). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Bioenergetic Disruption Case Study: Cyanide (CN^-) and Carbon Monoxide (CO) bind irreversibly to Complex IV (Cytochrome c oxidase) of the mitochondrial ETC, blocking electron transfer to O_2, halting proton pumping, collapsing the H^+ gradient across the inner mitochondrial membrane, and arresting ATP Synthase oxidative phosphorylation. Conversely, 2,4-Dinitrophenol (DNP) and thermogenin (UCP1) make the inner membrane permeable to H^+, uncoupling electron transport from ATP synthesis and dissipating proton-motive force as heat.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 3.5 to prove why the claim holds.

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